P530/1
BIOLOGY
(Theory)
Paper 1
Jul./Aug. 2026
3 hours

WAKISO-KAMPALA TEACHERS' ASSOCIATION (WAKATA)

WAKATA MOCK EXAMINATIONS 2026

Uganda Advanced Certificate of Education

BIOLOGY

Theory

Paper 1

3 hours

AKATA 2026

INSTRUCTIONS TO CANDIDATES:

This paper consists of two sections: **A** *and* **B**. *It has six examination items.*

Section **A** *has two compulsory items.*

Section **B** *has* **two** *Parts:* **I** *and* **II**. *Respond to one item from each part.*

Respond to **four** *items in total.*

All responses should be written in the answer booklet provided.

Any additional items responded to will not be scored.

FOR SCORERS' USE ONLY

ITEMS	SCORE	SCORERS' INITIALS
SECTION A		
1		
2		
SECTION B		
4/5		
5/6		
TOTAL SCORE		

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Turn

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SECTION A

Answer **all** *the items in this section.*

Item 1.

A plant biotechnology research centre developed a genetically modified tomato varie intended to improve resistance to environmental stress. Researchers introduced a recombina DNA sequence into the tomato genome leading to expression of a novel protein. Although th modification was expected to improve plant performance, some modified plants exhibited po growth and reduced vigour compared with the original variety.

Scientists investigated whether expression of the introduced gene had disrupted normal cellu processes involved in growth and development.

Table 1: Observations from Tomato Plants

Observation	Normal Tomato Variety	Modified Tomato Variety
Amount of newly detected protein	Regulated levels	Excessively high levels
Condition of mitochondria in leaf cells	Numerous intact cristae	Reduced number of cristae with structural damage
Activity of tonoplast transport proteins	Normal	Reduced
Rate of cell division in growing regions	High	Low
Overall plant growth	Normal development	Stunted growth

The researchers concluded that the reduced growth originated from cellular changes associated with expression of the introduced DNA sequence.

Task:

(a) Analyse how expression of the introduced recombinant DNA sequence affected cellular structures and metabolic processes, leading to reduced growth and development of the modified tomato plants.

(b) Propose measures that researchers and regulatory institutions could implement to improve the genetic modification process and ensure the responsible application of genetically modified plants. For each measure, provide a biological justification and consider ethical, ecological, or social implications.

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Item 2.

A crop improvement programme in the drought-prone cattle corridor of Uganda introduced two forage crop varieties for livestock feed production. During prolonged periods of high temperature and limited rainfall, farmers observed that Variety A remained green and continued growth, whereas Variety B showed severe wilting, premature leaf ageing and reduced biomass production despite receiving similar field management practices.

Plant physiologists investigated the physiological adaptations of the two varieties under the prevailing environmental conditions.

Table 2: Physiological characteristics of the two forage crop varieties

Parameter assessed	Variety A	Variety B
First carbon fixation enzyme	PEP carboxylase (PEPC)	RuBisCO
Root system development	Deep and extensive	Shallow and less extensive
Stomatal response during water stress	Partial closure	Prolonged closure
ABA concentration during water stress	Moderate and regulated	High
Cytokinin concentration during water stress	Maintained	Reduced
Biomass production	High	Low

The researchers concluded that differences in physiological adaptation and hormon regulation influenced the productivity of the two crop varieties under environmental stress.

Task:

(a) Analyse how differences in carbon fixation, root development, stomatal regulation and hormonal interactions influenced the growth and productivity of the two crop variet under the prevailing environmental conditions.

(b) Propose sustainable agricultural practices that farmers in drought-prone regions could adopt to improve crop productivity under similar environmental conditions, and justif how each practice enhances plant survival and productivity.

Irrigation

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SECTION B

Part I

Answer only **one** *item from this section*

Item 3.

A fish farmer in western Uganda noticed that fish in one of his ponds became less active, showed delayed responses to stimuli, and exhibited increased signs of disease after a nearby agro-processing facility discharged untreated wastewater into a stream supplying the pond.

A fisheries scientist investigated how changes in the aquatic environment affected the physiological functioning of the fish. Samples were collected from fish in an unaffected pond and the affected pond for further investigations.

Table 3: Physiological Observations of Fish from Two Ponds

Parameter assessed	Fish from unaffected pond	Fish from affected pond
Dissolved oxygen in water (mg L^{-1})	7.8	2.6
Blood oxygen saturation (%)	94	61
Heart rate (beats min^{-1})	62	108
Response time to external stimulus (s)	0.4	1.5
White blood cell count ($\times 10^6$ mL^{-1})	8.2	17.4

Further investigation showed that the waste water reduced oxygen availability in the pond and introduced substances that stressed the fish. The scientist explained that survival in changing environments depends on effective coordination between respiratory, circulatory, nervous, and immune systems.

Task:

(a) Analyse how the environmental changes affected oxygen transport, circulatory regulation, nervous coordination and immune responses in fish from the affected pond.

(b) Propose sustainable strategies that fish farmers, local communities ~~communities~~ and industries could implement to prevent water pollution and physiological disruptions in aquatic organisms and improve aquatic animal health, productivity and welfare. For each strategy, Justify how it contributes to these outcomes.

anerabically.
anerobic

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Item 4.

A poultry farmer in eastern Uganda introduced a new housing system intended to improve egg production. During a period of unusually high environmental temperatures, the farmer observed that some birds became weak, showed reduced feed intake, produced fewer eggs, and developed frequent infections.

A veterinary officer investigated the physiological responses of two groups of birds kept under different management conditions.

Table 4: Physiological Observations in Two Groups of Birds

Parameter assessed	Group A (Well adapted)	Group B (Affected)
Core body temperature (°C)	41.0	43.5
Plasma ADH concentration (pg mL^{-1})	28.4	9.7
Urine osmotic concentration (mOsm kg^{-1})	1240	520
Antibody concentration after vaccination (arbitrary units)	86	34
Disease incidence during observation period (%)	8	41

Further investigation showed that Group B had inadequate access to drinking water and poor management practices during periods of high environmental temperature. The veterinary officer explained that maintaining survival during heat stress requires coordination between the nervous system, endocrine system, kidneys, circulatory system, and immune system.

Task:

(a) Analyse how disruption of physiological regulation affected temperature control, water balance and immune function in Group B birds.

(b) Propose sustainable management strategies that poultry farmers could implement to maintain homeostasis, improve productivity and promote animal welfare during periods of high environmental temperature. For each strategy, justify how it supports physiological stability and performance.

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Part II

Answer only **one** *item from this section*

Item 5.

A bean-growing cooperative in eastern Uganda relied on a pyrethroid pesticide to control black bean aphids (*Aphis fabae*). During the first few years, crop damage was minimal and yields improved. However, after repeated use of the same pesticide, farmers observed increasing aphid infestations despite applying more pesticide. Field observations also showed that ladybird beetles preferentially fed on black aphids because they were more visible on green bean leaves.

Researchers found that pesticide resistance was controlled by a dominant allele (R), while body colour was controlled by a separate gene (B for black, dominant over b for brown). To determine whether the two genes were inherited independently, a heterozygous resistant-black aphid was test-crossed with a susceptible-brown aphid.

Table 1: Offspring from the Test Cross

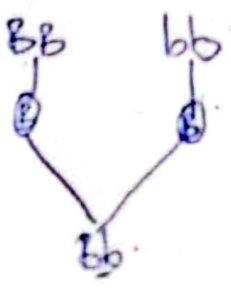
Phenotype	Number of offspring
Resistant, black	208
Resistant, brown	42
Susceptible, black	46
Susceptible, brown	204

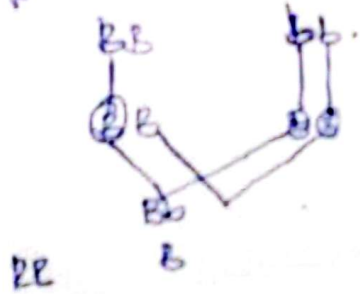
The researchers suspected that the genes controlling pesticide resistance and body colour were located on the same chromosome.

Task:

(a) Analyse how inheritance, linkage, pesticide selection and predation influenced the genetic composition of the aphid population over time.

(b) Propose sustainable pest-management strategies that farmers could implement to reduce crop losses while slowing the evolution of pesticide resistance, and explain how each strategy contributes to these outcomes.





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Item 6.

A district in central Uganda has experienced extensive deforestation due to charcoal burning, agricultural expansion and settlement development. Environmental officers monitored changes in forest cover, atmospheric carbon dioxide concentration and crop productivity over a ten-year period.

Table 6: Environmental Changes in the District

Parameter assessed	2015	2020	2025
Forest cover (%)	74	58	39
Atmospheric carbon dioxide concentration (ppm)	410	432	458
Average annual rainfall (mm)	1450	1320	1180
Average crop yield (t ha^{-1})	3.8	3.2	2.4
Number of wild animal species commonly recorded	42	31	19

Researchers concluded that human activities were altering ecosystem balance and affecting both biodiversity and agricultural productivity.

Task:

(a) Analyse how the observed decline in forest cover influenced the carbon cycle, biodiversity, ecosystem stability and agricultural productivity.

(b) Propose sustainable strategies that communities and local governments could implement to reduce carbon emissions, conserve biodiversity and improve long-term food security. Justify how each strategy contributes to these outcomes

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